



Bipem Engineering & Consultancy Ltd.

Reg. & licensed Hydrologists, Water Engineers and EIA & Audit Experts.

COMPANY PROFILE

Prepared by:

Bipem Engineering and Consultancy Ltd.

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PROFILE SUMMARY:

Date of Incorporation:

9th April, 2015

PIN and VAT Reg. No.:

P051539461S

Postal Address:

P.O. Box 29489-00100,
Nairobi.

Email: bipemholdingsltd@gmail.com

Website: www.bipemengineering.co.ke

Physical Address:

2nd Floor, Rm L03, Doctors Plaza,
Ruiru.

Bankers:

Kenya Commercial Bank
Ruiru Branch

CATEGORY:

NCA 6

DUNS No. :

85-1759966

1.0 WHO WE ARE

Bipem Engineering and Consultancy Ltd is a fully registered and incorporated Kenyan company delivering homegrown water and environmental solutions since the year 2015.

Founded and incorporated in 2015 as Bipem Holdings Limited, we changed the name to Bipem Engineering and Consultancy Ltd in May 2021 to echo our mandate: **offering engineering solutions and consultancy**. We have over 7 years of experience in data science and analytics, GIS and Remote Sensing capacity building, climate change analysis, and Water, Sanitation and Hygiene (WASH), borehole drilling and equipping, pump installations, water pan construction and desilting and dam construction. We also undertake hydrogeological surveys, feasibility studies and environmental impact assessment (ESIAs).

We rely on state-of-the-art technology earth observation skills and technology to derive models and data to understand the nexus between climate change and water-cycle, changes and patterns. We possess a wealth of combined local knowledge across the data science, water, energy and environment sub-sectors and have recognized the benefit of a synergetic resource pool to address the normally pressing needs of globally established companies, looking to partner with and nurture local upcoming companies in the industry.

2.0 WHY US?

We are a registered and licensed water works and services contractor (NCA Category 6) with a pool of zealous, skilled and experienced engineers, hydrologists and EIA experts. Our expertise in data science, water resources development, civil engineering and construction, management, and environmental engineering is unrivalled.

We also offer construction equipment and lifting equipment hire and maintenance. Our fleet of forklifts, cranes and construction equipment is modern and operated by professional and experienced staff.

2.1 OUR VISION

To be the company of choice in the delivery of engineering solutions valuable to our customers.

2.2 OUR MISSION

Use science and technology to develop people-centered solutions in water, energy and environment for Africa.

2.3 OUR VALUES

- **Integrity:** Be consistent in our actions, values, methods, measures, principles, expectations, and outcomes and adhere to ethical principles and soundness of character and judgement in all that we do.

- **Engagement:** Involving our clients and stakeholders at every step of our undertakings to understand and exceed their expectations.
- **Due Diligence:** Supporting our clients to **identify** and **manage** operational bottlenecks that significantly impact CAPEX, create reputational risks, divert management resources and reduce asset values. We examine all issues, no matter how technical and complex, and report back in clear commercial terms. We then design and implement action plans to correct any concerns identified.
- **Growth:** We aspire to outperform the markets where we choose to work
- **Teamwork:** Harmonizing our unique skills enables us to best achieve the mission. To maintain a steadfast adherence to our code of ethics and corporate governance in all our dealings.

2.4 OUR PILLARS

3 deep-rooted values guide our decisions in pursuing our ambitions and interests:

- ✚ **People:** We recognize our projects and engagements are people-centered. So, to excel in any environment, our greatest strength is our employees' commitment to safety and customer satisfaction.
- ✚ **Technology:** Our commitment to technology and quality is the basis of our competitive edge.
- ✚ **Returns:** We adopt market-driven business models to create business footprints to anchor our future growth and independence. Unwavering dedication to our clients and their values.

2.5 OUR OBJECTIVES

- To be the company of choice in the safe and sound delivery of water projects
- To build partnerships for effective service delivery
- To strengthen the customer relations and corporate image



Supervision of Sereolipi & Ntabasi Borehole rehab works and reticulation improvement in Samburu East: ACTED, March-June 2020



Construction of VIP latrines in Ngilai in Baragoi, Samburu North: ACTED, July 2020

3.0 OUR SERVICES

Bipem offers a complete range of engineering solutions and services including design, project management and supervision, civil works and O&M; thanks to our experienced staff who understand the Kenyan business terrain and are committed to the successful delivery of projects where we work.

3.1. WATER SERVICES

We have unique experience and expertise in hydrology, integrated water resources management, groundwater development, the environment and climate change drawn from a pool of experienced professionals with more 50years combined experience and a deep understanding of the nexus between science and climate; critical to offering tailor-made solutions in:

- ✓ Water and Civil Works
- ✓ Borehole drilling, supervision & support
- ✓ Water use permit & conveyance
- ✓ Flood risk assessment
- ✓ Hydraulic structures design
- ✓ Ground water resources quality, modelling, planning and management
- ✓ Integrated water & environmental management

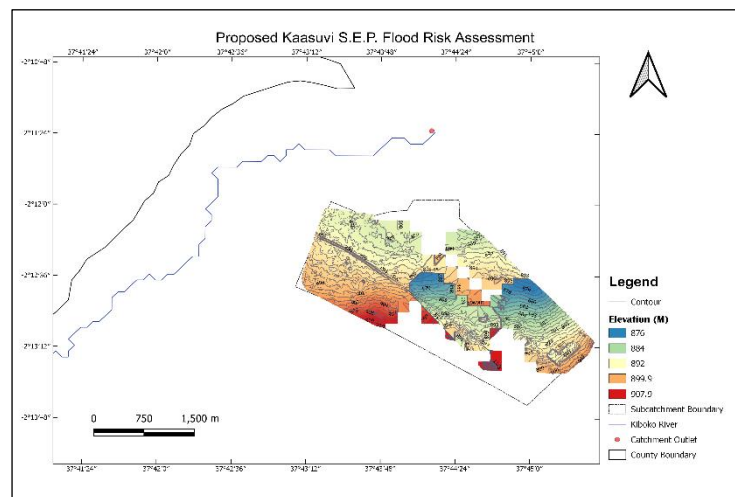


3.2. ENVIRONMENTAL SOLUTIONS

We offer consultancy in Environmental Management including the following:

- ✚ Environmental Impact Assessment (EIA)
- ✚ Environmental Audit
- ✚ Environmental Management Systems (ISO 14001)
- ✚ Application & renewal of NEMA licences/permits
- ✚ Water quality testing and analysis
- ✚ Noise Level measurement and analysis
- ✚ Participatory Rural Appraisal (Community Mobilization)

3.3. GEO-INFORMATION AND MAPPING SERVICES



We develop and maintain spatial databases and data analytics solutions to support prototyping and production of urban, regional, natural resource, business and utility databases. We also offer the following geospatial and mapping solutions:

- ✚ **3D/Drone Mapping and Remote Sensing:** Incorporating Drones, GIS 3D analyst, Google Earth as well as Sketch Up tools to develop:
 - 3D interactive models (DEMs & DSMs).
 - High Resolution Satellite & Drone imagery
 - GIS and remote sensing training
 - Map Design and Production Services
 - Land use/land cover maps
- ✚ Land, Road network, Engineering and Topographical Surveying.

3.4. MATERIAL HANDLING AND CONSTRUCTION EQUIPMENT LEASING, HIRE AND MAINTENANCE

We also offer a range of construction and lifting equipment. Our fleet is specialized, versatile and robust for both outdoor and indoor operations. We also enjoy access to a broad network that effectively links us to renowned industry players and other critical stakeholders. Below are some of our recent undertakings:

BIPEM IN ACTION:



Construction equipment lease to Unity Homes Tatu City. Inset: Phase 1 of the West wing ground breaking in 2018

Construction of Administration Block in Kithasyu Sec School- NG-CDF of Kibwezi East 2020/2021



Provision of heavy machinery and construction equipment in Ruiru Dam expansion works in Ng'ethu Rujawasco-2018/19



Provision of Crane to Schlumberger. Supporting Tullow Oil BV operations in Lokichar oilfields in Turkana-2016/2017



Excavation works Proposed residential flats in Kahawa Sukari
for Jamio Hldings Ltd- Jan, 2020



Tunneling of Nlogo-Kundu. Lifting services and trenching
Subcontractor

4.0 STATUTORY REGISTRATION AND PROFESSIONAL MEMBERSHIP

4.1. CERTIFICATE OF INCORPORATION

4.2. CR-12

4.3. PIN CERTIFICATE & TAX COMPLIANCE

4.4. US GOVERNMENT DUNS NUMBER

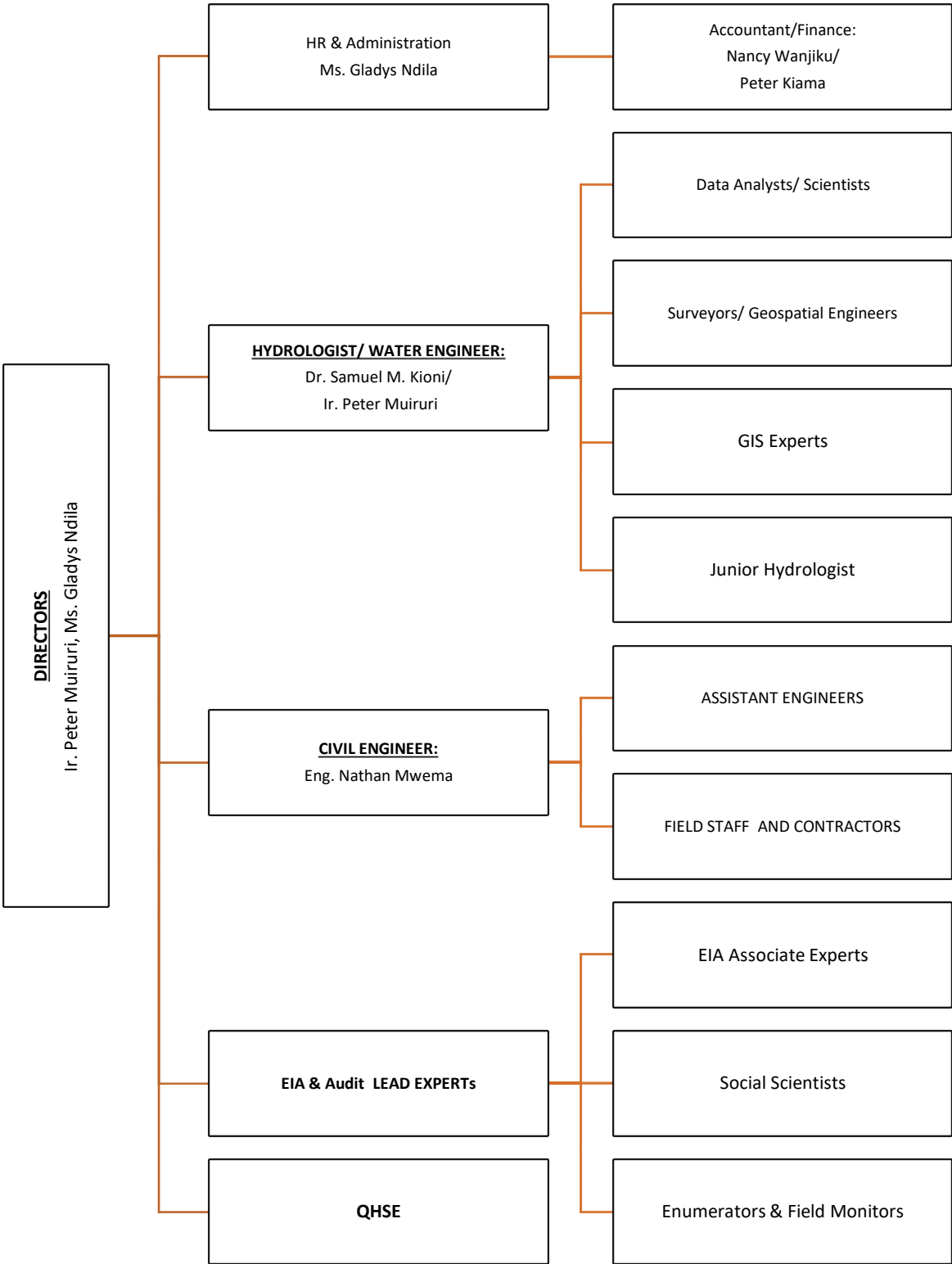
4.5. BUSINESS PERMIT/PROOF OF PHYSICAL OFFICE

4.6. NEMA CERTIFICATE AND LICENSE

4.7. NCA-6

4.8. LEGAL REPRESENTATIVE(S)

5.0 ORGANIZATION STRUCTURE



6.0 THE TEAM

6.1. DIRECTOR: PETER MUIRURI, MSc. IEK, HSK, GE

Peter is an experienced hydrologist, drilling and WASH expert with a diverse background spanning over 8 years in water resources development with a demonstrated history of working in the oil and gas industry and in the humanitarian sector. He holds a master's degree in Water Resources and Environmental Management focused on the application of GIS and remote sensing in water resources and environmental management from ITC, University of Twente, and a BSc degree in Water Engineering from Kenyatta University. He is a registered hydrologist, graduate engineer and an EIA & Audit Lead expert.

He is experienced in oilfield drilling and exploration, water resources development and management, emergency and post-emergency WASH programming, borehole drilling and prospecting (3D and 2D geophysical Resistivity surveys for geotechnical surveys, tomography, aquifer mapping and groundwater exploration) and environmental impact assessment.

He worked for Schlumberger as a drilling fluids engineer in the first phase of the Tullow Oil exploration and appraisal program in Kenya, and drilled on eight exploratory wells and developed five completion wells in the Turkana Oil Field; Agete-1, Ngamia-4, 5, 6, 7, 8, Ekales-1, Cheptuket-1, Badada-1 and Sala-3 in Wajir. He is conversant with various mud engineering applications and oil and gas tools and various drilling fluid formulations: water, oil and synthetic-based mud. He has managed and coordinated a spectrum of operations and field personnel and is versed in project planning, management, drilling fluids inventory management and HSE. He has worked for OXFAM, Feed the Children, private drilling water consultancies and water companies at different capacities. At Feed the Children, he was instrumental in rolling out the FTC livelihoods empowerment program in Samburu and Turkana counties and developed several donor-funding proposals.

Currently, he is involved in the design and construction of earth dams in Machakos County, and has a vast experience in the use of Civil 3D, ArcGIS, QGIS, MATLAB, ILWIS, and HEC-RAS from design and planning in the water sector. He specializes in the use of Models and GIS in the design of dams and is also involved in training. Peter is skilled in hydrology, GIS, project management, drilling fluids, inventory management, borehole drilling, gravity water systems, and PHAST/CHAST.

6.2. DIRECTOR: GLADYS NDILA

Gladys is a seasoned procurement officer with 10 years of experience in executive planning and production. Dedicated and committed to growing and expanding businesses and have a proven track record of success. She has extensive knowledge of business practices, economics, and acquisitions. Prioritize strong communication skills to delegate tasks and lead employees effectively. ***See Appendices for detailed CV.***

6.3. HYDROLOGIST/ENGINEER: DR. SAMUEL MUREITHI KIONI, PhD, HSK, EIK.

Dr. S.M. Kioni is a distinguished consulting hydrologist, engineer and lead environmental expert and a member of several professional bodies including:

- The Institution of Civil Engineers (ICE)- Member No. 81532768,
- Hydrological, Society of Kenya (HSK) - Member Number 00024/09/2001.
- International Association of Hydrological Sciences (IAHS)-Member Number 5969.
- Licensed Qualified Water Resources Professional (Hydrologist) No. WD/WRP/241 under Water Act.
- Licensed EIA/A Lead Expert No. 2672 under EMCA Act by NEMA, Member Environmental Institute of Kenya.
- Member of Association of France Alumni in Kenya (AFRAKEN). ***See Appendices for detailed CV.***

6.4. ENGINEER (CIVIL): NATHAN MWEMA BSC, PE ENG., MIEK.

Nathan holds a Bachelor of Technology, Civil and Structural Engineering, from Moi University; and a Diploma in Project Management, Kenya Institute of Management His other certificates include Computer Aided Design, CAD, MS Project Management and Microsoft Office.

He is registered professional engineer with the Engineers Board of Kenya, **Reg. No. A2729** and a member of the Institution of Engineers of Kenya (M.I.E.K).

He has practiced for the last 19 years; and his key works include;

- 1) Team Leader cum Hydraulic Engineer: Studies, design work, drawings, design reviews and preparation of bid documents for weir complete with off take structure for irrigation, water supply projects; for Coda And Partners (Africa) Ltd
- 2) Project Manager with Ambitious Construction Company Limited- Works co-ordination, Quality control and assurance, Time/ progress control, Cost control for Construction of Ngenge Farm Income Enhancement Project- Uganda
- 3) Production Manager: Studies, design work, drawings, design reviews and preparation of bid documents for water and sanitation for Warren Enterprises Limited
- 4) Associate Water Design Engineer: Review of test pumping reports and design of the appropriate pumps, storage tanks and associated water reticulation system (pipes) for small scale holder Irrigation schemes and domestic borehole water supplies for INSTA-PUMPS Engineering
- 5) Studies, design work, drawings, design reviews and preparation of bid documents for water and sanitation projects for
- 6) Site Agent: Works co-ordination, Quality control and assurance, Time/ progress control, Cost control for Construction supervision of Lower Sio Irrigation Scheme for Lee Construction Ltd.
- 7) Design Review and Construction Supervision for Rehabilitation and Augmentation for Oloitokitok Water and Sanitation Project
- 8) Design Review and Construction Supervision for Rehabilitation and Augmentation for Kiambere Mwingi Water and Sanitation Project.
- 9) Construction supervision for Water Supply Works in, Nairobi County. These are Mwengenyne, Matopeni, Maana, Spring Valley Estate, Kayole Soweto and Riverbank Settlements in Embakasi Sub County of Nairobi County.
- 10) Construction supervision for Water Supply Works in, Nairobi County. These are Mwengenyne, Matopeni, Maana, Spring Valley Estate, Kayole Soweto and Riverbank Settlements in Embakasi Sub County of Nairobi County.
- 11) Construction of Lower Sio Irrigation Project, Busia Kenya
- 12) Designing and Supervision of construction/ Improvement of water control structures of DASIP, Ministry of Agriculture and Co-operative, Republic of Tanzania
- 13) Feasibility Study, Detailed Design And Preparation Of Tender Documents For Upper Nzoia Irrigation Development Project
- 14) Designing and Supervision of construction/ Improvement of water control structures of DASIP, Ministry of Agriculture and Co-operative, Republic of Tanzania
- 15) Design Review and Bid Document for Ahero and West Kano Irrigation Development Project
- 16) Feasibility Study, Detailed Design and Bid Document for Nyatike Dam Development Project
- 17) Feasibility Study, Detailed Design and Bid Document for Soin - Koru Multipurpose Dam

6.5. GEOSPATIAL ENGINEER: ROSE MAYIENDA

Rose holds a degree in Geography and is a GIS and Remote Sensing expert with over 23 years extensive experience working in various fields including, population surveys, agriculture wildlife surveys, social-economic surveys, biodiversity conservation, land-use planning, sustainable development and environmental management projects. This GIS expertise sets her pan-African footprint in Somalia, Kenya, Tanzania, Democratic Republic of Congo, Rwanda, South Africa, Zambia, Zimbabwe, Botswana, Namibia, Burkina Faso, Cameroon and Nigeria.

Her areas of competence include:

- 1) GIS mapping system setup, implementation, data management and map building
- 2) In-depth knowledge of GPS technology, field data collection and GIS integration.
- 3) Excellent knowledge of GIS environment in African landscapes
- 4) Compiling geographic data from a variety of sources and media including field data, satellite imagery, aerial photographs, existing maps, secondary data
- 5) Analyzing and integrating spatial data using various tools and technologies
- 6) Analyzing spatial data for geographic statistics to incorporate into documents and reports
- 7) Disseminating GIS information in appropriate graphic formats for multiple audiences
- 8) Training material design and training staff, partners and community members on GIS and GPS use.
- 9) Experience in the use of assessment modeling tools like INVEST, Arc SWAT and statistical analysis tools like SPSS.
- 10) GIS software installation/troubleshooting
- 11) Website content management

Rose has worked with UNFPA, African Wildlife Foundation and Kenya Wildlife Service, to mention just but a few. She too has published in the area of remote sensing

6.6. SURVEYOR: LINUS KIHARA MWANGI

Linus Kihara is a hardworking, results-oriented, knowledgeable professional land surveyor with experience in cadastral surveying, engineering and topographic surveying. He has skills and knowledge in the use of: Geographic Information Systems, Geo-Statistics, Spatial Modelling, Global Positioning Systems, Aerial Photography, Remote Sensing and Information Technology.

He holds a Master of Science in Geo-Informatics from the University of Twente, the Netherlands, a Diploma in computer programming and a Bachelor of Science in Geomatic Engineering and Geospatial Information Systems from Jomo Kenyatta University of Agriculture and Technology.

He is trained in Global Navigation Satellite Systems (GNSS) data processing course, Photogrammetric training in DSS system. Direct georeferencing system and processing of aerial photography to develop orthophotos and final mosaic, the Lidar Scanner (Q680) and DSS camera systems integration for processing Lidar data from laser scanner and developing aerial photography and Use of Terrasolid suite to process Lidar data from raw files to finished products such as Digital Elevation Models, profiles and contour generation.

As a practicing surveyor, some of his works involved; Topographical survey of various courts in Kenya, Mocha Holding Residential Development – Limuru. Subdivision planning and Setting out of over 50 housing units; Rift Valley Railways - Changamwe, Makadara and Kampala, Topographical Survey of the respective Marshalling Yards. Kimwarer and Aror multi-purpose dams - Elgeyo Marakwet and Extension of geodetic control, aerial mapping and topographic survey of the proposed dam.

While working for the Ministry of Lands he successfully completed projects in;

- 1) Creation of Land Information Systems in Nairobi and other major towns.
- 2) Development of the Kenya Reference Frame including GPS observations and data processing.
- 3) Building the new Kenyan Geodetic Control Network.
- 4) Proposed Lamu Port Preliminary perimeter survey.

- 5) Cadastral survey of parcels for the resettlement of Internally Displaced Persons (IDP)
- 6) Kenya - Ethiopia and Kenya-Tanzania Boundary reestablishment survey.

While working for Geomaps Ltd, he led the following projects

- 1) Topographical and cadastral survey and creation of a Land Information System (LIS) for the Kenya -Uganda oil pipeline extension. (From Eldoret to Kampala)
- 2) Creation of a GIS database for the Kenya National Bureau of Statistics (KNBS) for use in the general census
- 3) Cadastral survey and creation of GIS/LIS database for the surveyed Government lands in Uganda
- 4) Engineering and topographic survey using GPS receivers in Soroti-Moroto road (150km) and Muyembe-Moroto road (195 km) Uganda
- 5) Aerial photography, Topographic mapping, Lidar mapping and creation of a geodatabase for the Lamu Port South Sudan Ethiopia Transport corridor (LAPSSET)
- 6) Aerial photography, GPS surveys, Lidar mapping and creation of a geo-database for Kenya electricity and Transmission Company limited (KETRACO)
- 7) Setting out of Olkaria IV geothermal station including levelling and creating profiles for the Kenya Electricity Generating Company (KENGEN)

6.7. GIS AND REMOTE SENSING EXPERT/TRAINER: WILLIAM MAKAU

1. **Name of staff :** William Makau
2. **Nationality:** Kenyan
3. **Education:**

<u>School, college and/ University Attended</u>	<u>Degree/certificate or other specialized education obtained</u>	<u>Date Obtained</u>
University of Twente, ITC, Netherlands	MSc Geo-information Science and Earth Observation in WREM	2014-2016
University of Nairobi, Kenya	BSc Environmental and Bio-systems Engineering	2008-2013

4. **Professional Association:** Graduate Engineer with Engineers Board of Kenya (EBK)
Environmental and Social impact assessment Associate expert (NEMA)
5. **Other Relevant trainings:**

Certificate in Risk analysis and Management in projects –KSG (Aug 2021)

Academic Awards

- Young African Leaders Initiative Awards- Certificate in Management strategies for people and Resources (March 2019)
- Young African Leaders Initiative Awards- Certificate in Understanding and adapting to Climate change (March 2019)
- Netherlands Government Scholarship for Post Graduate studies (NFP-NUFFIC)

7. Countries of Work Experience:

8. Languages:

Language

Reading

Writing

Speaking

English	Excellent	Excellent	Excellent
Kiswahili	Excellent	Excellent	Excellent

9. Employment Record:

2014 – Present Water Engineer, Machakos County Government

Responsible for project identification, feasibility studies, design, development of Bills of Quantities, and implementation to solve the water scarcity menace. Data collection, analysis and reporting on project status and general sustainability analysis for projects. Development of proposals for funding and management strategies for hydrological projects and infrastructure.

June 2020 – Present Associate Consultant, Bipem Engineering and Consultancy

Implementation of water and WASH Engineering projects. Carrying out of surveys and designs for water projects in the ASAL regions of the country. Carrying out of Climate smart agriculture guided trainings and capacity building for farmers and project beneficiaries.

Use of GIS and RS technology and applications to develop and analyse hydrological products for problem solving. Training on new technologies (UAVs and RS) for surveys and change detection in Agriculture and Engineering for problem targeted solutions.

May 2013- June 2014 Ocra Engineering Consultants- Nairobi

Responsible for field surveys and reporting on project status, Monitoring and evaluation of water and irrigation projects. Proposal development and presentation to funding entities.

10. Detailed Tasks	12. Work Undertaken Illustrating Capacity and Competence
Project surveys, design and implementation	Name of assignment or project: Kenya Climate Smart Agriculture Program: Providing climate guided agricultural solutions to small scale farmers in select wards of Machakos county. Funded by the world Bank Year: 2019-2023 Location: Kenya Client: Machakos County Government for World Bank Main project features: Carrying out project surveys, designs, development of the bills of quantities and overseeing the implementation of the projects. Position held: Project Engineer
Coordinating overall project implementation and partners, training, Remote sensing and data analysis, Report writing	Name of assignment or project: Small scale Irrigation and Value addition program. Improving the irrigation potential of the selected water deficient wards in Machakos County through construction of new and rehabilitation of old dilapidated irrigation infrastructure Year: 2019-2022 Location: Kenya Client: Machakos County Government Main project features: Carrying out project surveys, designs, development of the bills of quantities and overseeing the implementation of the projects. Positions held: Surveyor

	Name of assignment or project: Cushioning Pastoral communities from droughts through construction of water pans and sinking of Boreholes in Samburu and Baringo Counties. Funded by ACTED Kenya. Year: 2020 Location: Kenya Client: ACTED Kenya Main project features: Overseeing the implementation of Ntabasi and Sereolipi borehole rehabilitation, solar powering and reticulation. Positions held: Project Engineer
Data collection, analysis and reporting	Name of assignment or project: Machakos County Government Water Mapping aimed at establishing the water resource status of the county for informed resource distribution. Year: 2015 Location: Machakos County Client: Machakos County Government, Ministry of Water, irrigation and Sanitation. Main project features: Geological mapping and identification of all water resources in the county. Data analysis and reporting.
Training farmers, Water Resources and irrigation water resource management	Name of assignment or project: Irrigation Water Resource users training Year: 2021-2022 Location: Machakos, Kenya Client: Ministry of Agriculture, Kenya Main project features: Training farmers on proper management and sustainability of irrigation water resources. Positions held: Trainer

6.8. GIS AND REMOTE SENSING ANALYST: BII CHERUIYOT VINCENT

1. **Name of Staff:** BII CHERUIYOT VINCENT
2. **Proposed Position:** GIS and Remote Sensing Analyst
3. **Nationality:** Kenyan
4. **Education:**

<u>School, college and/or University Attended</u>	<u>Degree/certificate or other specialized education obtained</u>	<u>Date Obtained</u>
University of Twente, ITC, Netherlands	MSc Geo-information Science and Earth Observation for Applied Earth Sciences	2016-2018
South Eastern Kenya University, Kenya	BSc Hydrology and Water Resources Science (Hons)	2011-2015
KCA University	Certified Public Accountant (CPA – Sec 2, part 3)	2010-2011

5. **Professional Association:** Member International society for Photogrammetry and Remote sensing (ISPRS)

6. **Other Relevant trainings:**

Certificate in Spatial Data Science: The New Frontier in Analytics (ESRI)

Certificate in Imagery applications in GIS (ESRI)

7. Countries of Work Experience: Kenya, Ethiopia, Somalia, Uganda, Netherlands

8. Languages:

Language	Reading	Speaking	Writing
English	Excellent	Excellent	Excellent
Kiswahili	Excellent	Excellent	Excellent

9. Employment Record:

2019 – Present GIS & Remote Sensing Analyst, BIPEM Engineering and Consultancy

Responsible for research and support on development of solutions targeted towards environmental and social impact through leveraging our analytical and data capabilities. I provide oversight and coordinate activities under the technical, data and analytics departments.

April 2021 – Present ArcGIS Developer, Strukton Rail Nederland

Develop a Geo-Infrastructure database for processed Rail object Data Analyzed in ArcGIS pro and Desktop and visualized in ArcGIS enterprise.
Automation in FME to sync updates and assets upon change in attributes or addition or deletion of new Assets, between the database and work planning documents.
Maintenance activities planning and schedule in ArcGIS and Tableau.
Data integration for cross department use and silo breaking through development of a Geodata Infrastructure Architecture.

Sep 2020 – Present Volunteer (GIS&Data Team), 510 Global

Data driven document visualization data model for the Red Cross who is who platform Image sharpening and derive land cover land use maps using high resolution imagery for Haiti for change detection and hazard assessment.
Household survey data analysis for UNICEF Southern Africa region in response to covid19 situation and mitigation at local level.
Research on retrieval of big data e.g., rainfall data from ECMWF and satellite data in preparation for a PHD proposal.

Sep 2019 – Jul 2020 Volunteer (GIS&Remote Sensing), Kenya Red Cross, Nairobi, KE.

Developed automated scripts in Google Earth Engine using SAR data/GIS operations/Drone imagery and field observations to quantify impact to food security by Desert Locust invasion.
Developed workflows for automatic hazard (Floods, Food security, Landslide, and Drought) assessment and vulnerability analysis for efficient disaster response using R and Python languages and Google Earth Engine API.
Data processing and manipulation in R and Python for suitability and predictive analysis and visualization.
Generated digital nature interactive maps and storytelling maps or maps in general for visualization of data in ArcGIS and QGIS platform.

11. Detailed Tasks

12. Work Undertaken Illustrating Capacity and Competence

Coordinating and planning of Mapathons and mapping parties.	Name of assignment or project: Disaster Risk Reduction, Year: 2019 Location: Kenya Client: The Kenya Red cross Society. Main project features: Coordinating and conducting mapping parties to put communities at risk of disaster on the map for early warning early action Positions held: GIS Team lead.
Use of GIS and Remote sensing to map out the threat of Desert Locust Invasion to crop and animals in 16 counties.	Name of assignment or project: Assessing the impact of Desert Locust Invasion to crop and animals in 16 affected counties. Year: 2020 Location: Kenya Client: Food and Agricultural Organization. Main project features: Use satellite and drone imagery to determine the changes to vegetation cover after and before a desert locust invasion. Positions held: Remote Sensing Analyst.

6.9. EIA EXPERT: PHILIP JUMA BARASA

Phillip Jima is a hardworking, results-oriented, knowledgeable professional environmental scientist experienced in conduct of Environmental Impact Assessment/Audits, conducting occupational health and safety audits and conducting training in environmental management since 2004. He is a registered and licensed Lead Environmental Impact Assessment /Audit expert with the National Environmental Management Authority (NEMA) - NEMA Reg. No. 1857.

He holds a Bachelor's of Science in Environmental Science from Egerton University and a post-graduate diploma in community development and project management form Kenya College of Commerce and Hospitality.

Phillip has mentored and pupiled many including Bipem Engineering and Consultancy Ltd director Peter Muiruri during his time as the principal consultant for Eco Insight Environ Consult (Africa). Some of his works include:

Organization/Contact person	Project title	NEMA Ref.	Date of contract	Contacts
Nyali Estate Developers	Proposed flats along Links Road in Nyali Estae, Mombasa	PR/1/1/0596	September 2021	Mr. Hussein 0722753335
Vaalco Petroleum Ltd	Proposed Petrol Station along Ngong Road	EIA/5/2/540	February 2019	Arch. Raymond Wanjau 0722702735
Kidgrove Ltd/Mr. Korane	Proposed office block in Kilimani, Nairobi	EIA/5/2/452	March 2016	0724383762
Mr. Mahmood Ali Farah	Proposed flats in Eastleigh, Nairobi	PR/5/1/0413	April 2010	C/o Dr. Issa, 0724714593
Ahmed Shariff Ali Othman	Proposed shops & flats in Eastleigh, Nairobi	PR/5/2/0414	March 09	C/o Dr. Issa, 0724714593
Said Ali kaitany	Proposed Kolol Recreation Centre, Eldoret	RVP/PR/0119	May 09	0723270047
Neo Homes Ltd	Proposed town houses	PR/5/1/0571	June 09	C/o Mr. Yasir 0724900939

Mr. Salih Abdullah Mohamed	Proposed flats in Ngara, Nairobi	PR/5/1/1119	October 09	0722103809
Shemsedin Mohamed Siraj	Proposed shops & flats in Eastleigh, Nairobi.	PR/5/1/1223	November 09	C/o Arch. Raymond Wanjau 0722702735
Abdi Mohammed	Proposed shops & flats in Eastleigh, Nairobi.	PR/5/1/1328	November 09	0727775077
Rosewa Agencies Ltd	EIA for proposed maisonettes in Juja area	PR/5/2/5857	February 2009	020 - 6763277
Whitefield Place Ltd	Proposed lift installation in West lands, Nairobi	PR/5/2/6260	September 09	020-4454301/2/3
Central Bank of Kenya	EIA for the proposed residential house in Mombasa	CP/PR/0044	January 2009	020-863000/2861000

6.10. DATA ANALYST/GIS AND REMOTE SENSING TRAINER: DANIEL KILUU

- Name of Staff:** Daniel Kiluu
- Proposed Position:** Data Analyst
- Nationality:** Kenyan
- Education:**

<u>School, college and/or University Attended</u>	<u>Degree/certificate or other specialized education obtained</u>	<u>Date Obtained</u>
University of Twente, ITC, Netherlands	MSc Geoinformation Science and Earth Observation	2015-2017
Kenyatta University, Kenya	BSc Water Engineering (Hons)	2006 – 2011

- Professional Association:** Graduate Engineer – Engineers Board of Kenya (EBK)
- Other Relevant trainings:**

Certificate on Drinking water technologies for African countries (Anghui Institute of Business studies – China)

Certificate in Monitoring and Evaluation (Kenya School of Government)

Certificate in Social Risk Management (Kenya School of Government)

Academic Awards

 - Netherlands Government Scholarship for Post Graduate studies (NFP-NUFFIC)
- Countries of Work Experience:** Kenya
- Languages:**

Language	Reading	Writing	Speaking
English	Excellent	Excellent	Excellent
Kiswahili	Excellent	Excellent	Excellent

9. Employment Record:

2019 – Present Data Analyst, Bipem Engineering and Consultancy Ltd

Responsible for research and support on development of solutions targeted towards environmental and social impact through leveraging our analytical and data capabilities. I provide oversight and coordinate activities under the technical, remote sensing data and analytics departments.

April.2020 – Present Ag. Director Climate Change, Machakos County Government

Responsible for researching and supporting development of data responsive climate risk solutions including participatory vulnerability risk assessment, disaster risk profiling and data analysis for proposal development. In addition, I help the county develop key policy documents and institutional framework to access climate financing from various entities.

- April 2014 – Present** Senior Superintending Engineer – Surface Water, Machakos County Government
Responsible for designing of water structures across the county, including the use of Remote sensing data for hydrological analysis, generation of hydrological reports and technical reports. To achieve this we have employed the use of GIS data, Drone surveys and rainfall data for decision making.
- Mar 2017 – May.2018** GIS & RS trainer, Egerton University.
Responsible for development of geospatial training materials especially on modelling of irrigation efficiency using satellite data. Training Master’s students on how to use GIS and remote sensing for research and how to access and process various remote sensing products.
- Sep. 2012 – April. 2014** Part Time Lecturer, Kenya Water Insitute.
Responsible for developing training materials for various courses, and training on Hydrology, and introduction to Remote Sensing.

11. Detailed Tasks	12. Work Undertaken Illustrating Capacity and Competence
Coordinating the implementation of CRAL component in KCEP-CRAL	Name of assignment or project: KCEP-CRAL: Climate risk Year: 2021 Location: Kenya Client: County Government of Machakos/NDMA Main project features: Coordinating overall project implementation and partners, Training, Remote sensing and data analysis, Report writing and presentation Positions held: Ag. Director – Climate Change
Coordinating overall mapping, design and implementation of surface water projects	Name of assignment or project: Design and construction of surface water structures Year: 2014-present Location: Kenya Client: Machakos County Government Main project features: Coordinating overall project implementation and partners, Remote sensing and data analysis, Report writing and permit applications. Positions held: Senior Superintending Engineer – Surface Water
Training on GIS and Remote sensing	Name of assignment or project: Training on GIS and Remote Sensing for research on climate data analysis Year: 2017 Location: Kenya Client: Centre for Integrated Water and Basin Management (CIWAB) Main project features: Geographic Information System & Remote Sensing training, development of research materials and capacity building of MSc. Students.
Training and capacity development on the GIS field for KenGen	Name of assignment/Job or project: GIS installation & setup for KenGen staff Year: 2018 Client: KenGen Position held: Trainer Main Project Features: Supply and Installation of GIS software and training and remote sensing for monitoring of catchments and management of hydroelectric dams. Activities Performed: Training and capacity development on the GIS field. Creation of a GIS database, catchment delineation, and sediment transport modelling and capacity building on GIS and remote sensing.

7.0 OUR CLIENTS/ EXPERIENCE

We are customer conscious and committed to providing customized market-driven solutions anchored to our specialist expertise and in compliance with statutory legislation. Some of clients and experience include:

7.1. NATIONAL GOVERNMENT CDF (NG-CDF KIBWEZI EAST CONSTITUENCY) 2020/21

1. Construction of 2 classrooms block at Nguumo Boys Secondary School-Ministry of Education, Kibwezi East- **KES 2,340,330/- June 2021-Completion certificate attached**



2. Renovation of two classrooms at Kaliani Primary School for the sum of **KES 1,193,855.20- tender award letter attached**



3. Construction of Mikiyuni Assistant Chief Office at Mikuyuni Ass. Chief's office, Kibwezi, -**KES 1,190,770/- tender award letter attached**



4. Construction of an Administration block at Kithasyu Secondary School -**KES 3,000,000/-** *tender award letter attached*



7.2. COUNTY GOVERNMENT OF SAMBURU

Date: (Financial year 2019/2020) - May 2020

Construction of 2 classrooms at Ngilai Primary School, Samburu North-**KES 2,705,775/-**



7.3. OXFAM

Date: Dec 2019

Construction of Generator house (1 No.), construction of water trough (2 No.), construction of elevated tank tower (1 No.), supply and construction of mini-water kiosk (1 No.), supply and erection of chain link fencing and gate (1 No.) and all plumbing works at Shidley, Wajir West.



7.4. UNITY WEST LTD (UNITY HOMES TATU CITY)

Date: Dec 2018- June 2020

5. Supply of building materials and construction equipment (Excavator, backhoe, concrete mixers and motor grader)



6. Supply of crane and forklift services
7. Cabro paving of parking lot and walkways



7.5. AGRICULTURE AND CLIMATE RISK ENTERPRISE LTD (ACRE AFRICA)



Date: 2019 – 2020

Project Title: Agro-ecological zoning using GIS for climate vulnerable communities in Kitui, Kenya.

Project Objective: To develop Agro ecological zones for World Bank funded KCSAP using GIS and remote sensing technologies

7.6. POWER CHINA HUADON ENGINEERING CORPORATION

Date: Aug-Nov 2020: **Project:** Hydrology and hydraulics study for flood risk assessment for proposed solar power plant at Kaasuvi, Makindu.

7.7. FARMER'S CHOICE KAHAWA

Date: 2020 – current

Supply and lease of 3ton forklift



7.8. JIKO KOA

Date:

August 2019- December 2019

Supply of crane and forklift services



8.0 OUR PARTNERS

Partnerships lie at the root of all the positive transformation that we create. Together, we work on program development and delivery, research, and advocacy. Our partners have worked with us to deliver ever more innovative, effective and successful projects and programs and their support has been vital in achieving our Mission and Vision.

8.1 KYELENGE GENERAL CONTRACTORS LTD

Kyelenge General Contractors Ltd is a Kyelenge General Contractors Ltd is a fully incorporated construction and engineering firm in the Republic of Kenya registered by the National Construction Authority under category of NCA 2 for building works, water works as and road works. They provide a comprehensive building service which is delivered through a regional teams to national and international clients operating across a wide range of markets.

Kyelenge leases various construction equipment to Bipem Engineering and Consultancy Limited from time to time:

1. Graders,
2. Bulldozers
3. Tippers
4. Drum vibrators and concrete mixers etc.

8.2 CONSTRUCTION EQUIPMENT LEASE AGREEMENT

9.0 HEALTH, SAFETY AND ENVIRONMENT (HSE)

Our HSE Culture is based on bringing no harm to people, property and the environment. We seek to be the best in what we do, guided by our mantra: **Work Safe. Be Safe!**

9.1 MANAGING HSE

Quality, Health, Safety and Environment (QHSE) is a priority area for us. We foster the culture through HSE controlled processes and continuously strive to provide a risk-free working environment for our employees, partners and customers. The standards we uphold, and enforce conform to applicable legal and voluntary requirements to ensure are compliant with statutory requirements and observe best practice. HSE tenets and deliverables are subject to individual commitment by each team member, integrated in our performance management reviews and driven by senior management.

9.2 HEALTH SAFETY AND ENVIRONMENT POLICY

We are committed to conducting our businesses in a sustainable manner that guarantees the health and safety of our staff, contractors, clients, visitors and people. We strive to continually improve by using a combination of proactive and reactive monitoring techniques from which progress towards those objectives are be measured. We encourage a “no blame” culture with regards to HSE reporting. All reported HSE issues are duly investigated to identify the causes and prevent re-occurrence.

We aim to:

- ✓ Provide and maintain a safe, secure and healthy working environment.
- ✓ Comply with all applicable legal HSE requirements whilst developing and maintaining appropriate emergency response plans.
- ✓ Conduct ourselves safely to prevent accidents, incidents and environmental harm.
- ✓ Encourage a preventative action culture through pre-task risk assessments.
- ✓ Ensure strict adherence to security and safety guidelines while undertaking our operations.
- ✓ Actively encourage feedback for continuous improvement.
- ✓ Ensure all incidents are reported, recorded and appropriate corrective actions taken.
- ✓ Engage and collaborate with contractors and sub-contractors to ensure their health and safety capability and competence fulfill our expectations.
- ✓ Communicate this policy to all: our staff, public and contractors working for the company to make them aware of their individual HSE obligations.
- ✓ Adhere to processes, specifications and procedures to achieve Customer Satisfaction.

- ✓ Maintain service quality through proactive quality assurance and continuous improvement.

Our policy will be delivered by generating and embracing a culture that does not tolerate threats to health, safety nor our operating environment. This policy will be reviewed and amended, as and where necessary.

Signed:



Ir. Peter Muiruri
Technical Director

10.0 Annexes:

ANNEX 1: CV'S OF KEY PERSONNEL

1. Position Title:	Assistant Project Manager
2. Name of Staff:	Eng. Nathan Mwema
3. Date of Birth:	1972
4. Nationality:	Kenyan
5. Education:	1998 Bachelor of Technology, Civil and Structural Engineering, Moi University.
Other Training:	2018 Diploma in Project Management, Kenya Institute of Management Computer Aided Design, CAD, MS Project Management and Microsoft Office.
Professional Membership & Societies:	Registered Engineer, (Kenya Engineers Registration Board Serial No A2729) Member of the Institution of Engineers of Kenya (M.I.E.K)

6. EMPLOYMENT RECORD RELEVANT TO ASSIGNMENT: _

Period	Employing Organization and Title / Position (Contact Info for References)	Country	Summary of Activities Performed Relevant to the Assignment
May 2018 to date	Ambitious Construction Company Limited Assistant Project Manager <i>For Refs:</i> Director: Prashant Ramji Patel Tel: 256 414 273454 Email: ambitious@prayoshagroup.net	Uganda	Assistant Project Manager: Works co-ordination, Quality control and assurance, Time/ progress control, Cost control for Construction of Ngenge Farm Income Enhancement Project
Nov 2013 to April 2018	Runji & Partners Consulting Engineers Limited / Civil/Water & Sanitation Engineer <i>For Refs</i> Tel: +254 701 946020 Email: info@runji.co.ke Resident Engineer, Team Leader, Project Design Engineer: Eng. Runji Ngware	Kenya	Project Design Engineer: Studies, design work, drawings, design reviews and preparation of bid documents for irrigation, water supply and sanitation projects Works co-ordination, Quality control and assurance, Time/ progress control, Cost control for Construction supervision of several water and sanitation projects in Nairobi.

April 2013 To: Oct 2013	Coda And Partners (Africa) Ltd Team Leader, Water supply and Irrigation Engineer <i>For Refs:</i> E-mail: coda@codacorp.org Telephone: 254 20 2219052/312962 Mobile: 254 725 984064	Kenya	Team Leader cum Hydraulic Engineer: Studies, design work, drawings, design reviews and preparation of bid documents for weir complete with off take structure for irrigation, water supply projects
May 2012 To: April 2013	Lee Construction Ltd. Site Engineer: For Refs: Phone: 020-8099827/8/9 Email: admin@leeconstruction.co.ke	Kenya	Site Agent: Works co-ordination, Quality control and assurance, Time/ progress control, Cost control for Construction supervision of Lower Sio Irrigation Scheme
Period	Employing Organization and Title / Position (Contact Info for References)	Country	Summary of Activities Performed Relevant to the Assignment
2009	Coda And Partners (Africa) Ltd	Kenya	Team Leader: reconciliation the contractor's interim payment certificates and the monthly progress reports to prepare the contract draft completion reports for irrigation schemes
2010- 2012	Coda And Partners (Africa) Ltd - Ministry of Agriculture and Cooperative, Tanzania – DASIP Civil and Water Engineer	Tanzania	Team Leader: Studies reports, prepared and co-ordinate the field survey work, lead the design team in design of irrigation schemes and water supply facilities
2006 to 2008	Warren Enterprises Limited Civil & Structural Engineer/ Project Manager <i>For Refs:</i> Telephone: +254 020 3556838 Mobile: +254 736 550303	Kenya	Production Manager: Studies, design work, drawings, design reviews and preparation of bid documents for water and sanitation projects

1999 to: 2014	INSTA-PUMPS Engineering <i>For Refs:</i> Telephone: (+254) 020 206-6040, (+254) 722 804250 Email: info@insta-pumps.com		Associate Water Design Engineer: Review of test pumping reports and design of the appropriate pumps, storage tanks and associated water reticulation system (pipes) for small scale holder Irrigation schemes and domestic borehole water supplies.
1999 to: 2006	Viking Industries Ltd Civil & Structural Engineer/Project Manager <i>For Refs:</i> Telephone: (+254) 020 2528122 Email: jantech@jandu.biz	Kenya	Studies, design work, drawings, design reviews and preparation of bid documents for water and sanitation projects

8. Languages:

Language	Reading	Speaking	Writing
English	Good	Good	Good
Kiswahili	Good	Good	Good

9. Key Qualification: Adequacy for Assignment:

Reference to Prior Work/Assignments that Best Illustrates Capability to Handle the Assigned Tasks

Name of Assignment or Project: Design Review and Construction Supervision for Rehabilitation and Augmentation for Oloitokitok Water and Sanitation Project. Work entails:

Year: June 2015 **To** April 2018 **Location:** Oloitokitok, Kajiado County **Client:** Athi Water Services Board **Donor:** BADEA

Main Project Features:

- Detailed Design Review of water supply augmentation works, sewer works complete with waste water treatment works.
- Reconnaissance and confirmation of pipeline route
- Preparation of cost estimates, project scheduling, Drawings and Design Calculations, Tender Documents inclusive of Specifications of the Water Supply Works.
- Advice on implementation modalities.
- Preparation of pre-qualification criteria and evaluation of bidder.
- Review of the Contractors design for electro-mechanical (solar design, pumps and pipe fittings) proposals and installation.
- Project Monitoring, Evaluation, and Quality Control during works implementation.
- Impact Assessment for the water supply augmentation works. **Positions held:** Team Leader / Water Supply Design Engineer

Activities Performed:

- Review of the preliminaries design data: existing treatment system, water reticulation systems, proposed development plans and projections population,
- Design of a new sewer works complete with a waste water treatment works.
- Co-ordination of the topographic survey work.
- Undertaking design review for the detailed design work, preparation of detailed layout drawings and water pipe profiles, preparation of the material take-off, compilation of the bills of quantities and the tender documents.
- Preparation of the implementation criteria, pre-qualification criteria and evaluation of the bidder.
- Works supervision co-ordination (*Project Monitoring, Evaluation, and Quality Control*).
- Review of the Contractors design for electro-mechanical (solar design, pumps and pipe fittings) proposals and installation.

Name of Assignment or Project: Design Review and Construction Supervision for Rehabilitation and Augmentation for Kiambere Mwingi Water and Sanitation Project. Work entails:

Year: December 2016 *To* April 2018

Location: Mwingi, Kitui County

Client: Tanathi Water Services Board Donor: Italian Corporation Main Project Features:

These activities will include;

Review of the Contractors executive design sewerage works and the water supply works.

Review of the master list for tax exemption Positions held: Deputy Team Leader

Activities Performed:

Works supervision co-ordination (*Project Monitoring, Evaluation, and Quality Control*).

Review of the Contractors executive design for sewerage works and the water supply works.

Review of the master list for tax exemption.

Name of Assignment or Project: Construction supervision for Water Supply Works in, Nairobi County. These are Mwengenyi, Matopeni, Maana, Spring Valley Estate, Kayole Soweto and Riverbank Settlements in Embakasi Sub County of Nairobi County.

Year: May 2014 *To:* November 2015 Location: Nairobi County Kenya

Client:: Athi Water Service Board, Kenya

Donor: World Bank

Main Project Features:

The general scope of the works comprises the following:-

- Design review of water supply and sewer works;
- Project Monitoring, Evaluation, Quality Control of the works and making relevant and making recommendations to AWSB management for concurrence prior to accepting or implementing any change order.
- Implementation and review of the Environmental Management Plan.
- Provide expertise on construction sites as an Engineer.
- Monitoring quality and quantity of construction materials and equipment and verifying

against contract terms, inventory reports, and accepted construction standards.

- Setting out for the works.
- Laying of water supply.
- Installation of water fittings on an existing faulty water line (bulk meter, air valves, washout, pressure valves etc.);
- Diversion of drains and water ways.

Positions held: Resident Engineer

Activities Performed: Construction supervision of Water Supply contractors. This entails among others:

- Works co-ordination: site meeting organization, occurrence investigation, advice to client on cost implication of any cost-bearing occurrence and site activities diary documentation (equipment, personnel, weather and materials: both for availability and utilization).
- Quality control and quality assurance of the undertaken works.
- Time/ progress control.
- Cost control: verification of interim certificates, assessment of work instruction vis-à-vis cost implication, and notification to client of any additional cost, extension of time and claims.
- Inspection for practical completion.
- Progress review of the works.
- Onsite design review and recommendations to client on any changes required complete with cost implications.

Name of Assignment or Project: Construction supervision for Sewer Works in, Nairobi County. These are Matopeni, Maana, Spring Valley Estate, Kayole Soweto and Riverbank Settlements in Embakasi Sub County plus Ghetto and Mahira of Huruma, Starehe Sub-County of Nairobi County.

Year: May 2014 To: November 2015 Location: Nairobi County Kenya

Client:: Athi Water Service Board, Kenya

Donor: World Bank

Main Project Features:

The general scope of the works comprises the following:-

- Design review of water supply and sewer works;
- Project Monitoring, Evaluation, Quality Control of the works and making relevant and making recommendations to AWSB management for concurrence prior to accepting or implementing any change order.
- Liaison with World Bank Infrastructure Manager on any challenges and weekly progress of Kayole Soweto Project.
- Impact assessment for the Kayole Soweto Project.
- Implementation and review of the Environmental Management Plan.
- Provide expertise on construction sites as an Engineer.
- Monitoring quality and quantity of construction materials and equipment and verifying against contract terms, inventory reports, and accepted construction standards.
- Setting out for the works;
- Laying of water supply and sewer lines
- Casting of in-situ concrete Manholes and Masonry chambers.
- Laying of sewer lines and connecting them to existing trunk mains, and ancillary works;

- Installation of water fittings on an existing faulty water line (bulk meter, air valves, washout, pressure valves etc.);
- Diversion of drains and water ways.

Positions held: Resident Engineer

Activities Performed: Construction supervision of 4 projects. This entails among others:

- Works co-ordination: site meeting organization, occurrence investigation, advice to client on cost implication of any cost-bearing occurrence and site activities diary documentation (equipment, personnel, weather and materials: both for availability and utilization).
- Quality control and quality assurance of the undertaken works.
- Time/ progress control.
- Cost control: verification of interim certificates, assessment of work instruction vis-à-vis cost implication, and notification to client of any additional cost, extension of time and claims.
- Inspection for practical completion.
- Progress review of the works.
- Onsite design review and recommendations to client on any changes required complete with cost implications.

Name of Assignment or Project: Construction of Lower Sio Irrigation Project, Busia Kenya

Year: June 2012 **To:** May 2013

Client: National Irrigation Board, Kenya

Donor: Government of Kenya Positions held: Site Agent Location: Busia County, Kenya Main

Project Features:

- Weir Construction across Namanderema River and Sio River.
- Dam Construction across Munana River.
- Raiser pipe installation.
- Canals (main and secondary) complete with canal structures: drops, turn-outs, division boxes and checks structures.
- Technical designs, drawing and bid document preparation.

Activities Performed:

- Responsibility for study activities in the home office (Nairobi) and in the field,
- Co-ordinate all inputs by the key professional staff,
- Coordinate the preparation of all deliverables
- Responsible for undertaking
- Co-ordinate the setting out survey work, lead the technical team in reviewing the design work, drawings and preparation of working drawings for irrigation development: 5m high concrete water dam, intake complete with the associated irrigation infrastructures (drainage pipes, canals, canal structures and service road)
- Provide expertise on construction sites as a site agent.
- Monitor the Construction Sub Contractor's production or performance progress and notify subcontractor in writing of deficiencies observed during surveillance, and direct appropriate action to effect correction.
- Record and report to the Head Office and Resident Engineer incidents of gross faulty or nonconforming design that requires client attention, delays or problems.
- Monitoring quality and quantity of construction materials and equipment and verifying against contract terms, inventory reports, and accepted construction standards for each component

Name of Assignment or Project: Design Review and Construction Supervision for Gatundu Water and Sanitation Project. Work entails:

Year: July 2017 *To* April 2018

Location: Gatundu South, Kiambu County Client: Athi Water Services Board

Donor: African Development Bank

Main Project Features:

These activities will include;

- Detailed Design Review of three concrete dams, water treatment works complete with their bulk conveyance, water storage and distribution networks.
- Reconnaissance and confirmation of proposed dam sites, treatment works, storage location and the pipeline route.
- Preparation of preliminary designs, layout drawings and profiles, detailed designs drawings and schedules for the water supply works
- Preparation of cost estimates, project scheduling, construction Drawings and Design Calculations, Tender Documents inclusive of works Specifications of the Water Supply Works.
- Supporting community on work of with respect to contract deadlines and deliverables to ensure both community and contractor moves in the same speed for harmonic production and construction.
- Prepared of progress reports for work.

Name of Assignment or Project: Designing and Supervision of construction/ Improvement of water control structures of DASIP, Ministry of Agriculture and Co-operative, Republic of Tanzania

Year: 2010 *To:* 2012

Client: DASIP, Ministry of Agriculture and Co-operative, Republic of Tanzania

Donor: African Development Bank Positions held: Irrigation Engineer Location: Kigoma Region:-

Mkutu Irrigation Scheme in Kigoma Rural District, Main Project Features:

- Monitoring of works progress.
- Evaluation and recommendation of any site review required to facilitate work.
- Preparation of monthly progress reports.
- Financial control and Interim Payment Certificate review and approval.
- Compilation of operation and maintenance manual.
- Commissioning the project.

Activities Performed:

- Undertook construction supervision for Mkutu Irrigation Scheme in Kigoma Region for irrigation structures improvement and road construction.
- Advice on implementation modalities.
- Preparation of pre-qualification criteria and evaluation of bidder.

Positions held: Deputy Team Leader / Water Supply Design Engineer

Activities Performed:

- Review of the preliminaries design data: dams, water treatment system, water storage tanks, water reticulation systems, proposed development plans and projections population,
- Co-ordination of the topographic survey work.
- Undertaking design review for the detailed design work, preparation of detailed drawings and water pipe profiles, preparation of the material take-off, compilation of the bills of quantities and the tender documents.
- Preparation of the implementation criteria, pre-qualification criteria and evaluation of the bidder.
- Preparation of the implementation criteria, pre-qualification criteria and evaluation of the bidder.

Name of Assignment or Project: Feasibility Study, Detailed Design and Bid Document for Nyatike Dam Development Project

Year: November 2013 *To* April 2016 Client:: National Irrigation Board, Kenya Donor: Government of Kenya

Positions held: Team Leader/ Irrigation Engineer

Location: Migori County Kenya

Main Project Features:

- Feasibility Study for the project,
- Detailed design report and drawing for the project.
- Prepare Technical designs, drawing and bid document preparation.

Activities Performed:

- Responsibility for study activities in the home office (Nairobi) and in the field,
- Co-ordinate all inputs by the key professional staff,
- Coordinate the preparation of all deliverables.
- Responsible for undertaking feasibility study of the work to come up with:
 - design and drawings,
 - gross irrigation water assessment, design work,
 - drawings and preparation of bid document for irrigation development option: 48m high concrete face rock fill dam, water intake complete with the associated irrigation infrastructures (hydro- pump, pipes, pipe structures bulk conveyance canal and intake structure) complete with domestic water supply and elevated storage (for the Uriri and Migori areas) Nyatike Dam Development Project, Migori County, Republic of Kenya.

Name of Assignment or Project: Feasibility Study, Detailed Design and Bid Document for Soin - Koru Multipurpose Dam

Year: November 2013 *To* April 2017

Client:: National Water Conservation and Pipeline Corporation, Kenya

Donor: Government of Kenya

Positions held: Team Leader/ Irrigation Engineer

Location: Kisumu County Kenya

Main Project Features:

- Detailed design report and drawing for the project.
- Prepare Technical designs, drawing and bid document preparation.

Activities Performed:

- Responsibility for study activities in the home office (Nairobi) and in the field,
- Co-ordinate all inputs by the key professional staff,
- Coordinate the preparation of all deliverables
- Responsible for undertaking of gross irrigation water assessment,

Name of Assignment or Project: Feasibility Study, Detailed Design and Bid Document for Mirira Irrigation Development Project

Year: July 2013 **To:** Jan 2014

Client: Mirira Self Help Irrigation Group, Murang'a County, Kenya **Positions held:** Voluntary Design Engineer/ Project Coordinator **Donor:** Voluntary Work

Location: Murang'a County Kenya

Main Project Features:

- Feasibility Study for the project,
- Detailed design report and drawing for the project.
- Prepare Technical designs, drawing and bid document preparation.

Activities Performed:

- Responsibility for study activities in the home office (Nairobi) and in the field,
- Co-ordinate all inputs by the key professional staff,
- Coordinate the preparation of all deliverables
- Responsible for undertaking
- Undertaking feasibility study of the work to come up with design and drawings, lead the design experts in fields survey and assessment, design work, drawings and preparation of bid document for irrigation development option: weir, water intake complete with the associated irrigation infrastructures (pipes, pipe structures, fittings (bulk meter, air valves, washout, pressure valves etc.) Elevated Storage Tanks and intake structure) for the Mirira Self Help Irrigation Group, Murang'a County, Republic of Kenya.
- Facilitated the group in sourcing for financing with the National Irrigation Board and licenses with WRMA and NEMA.

Name of Assignment or Project: Design Review and Bid Document for Ahero and West Kano Irrigation Development Project

Year: June 2013 **To:** October 2013 **Client:** National Irrigation Board, Kenya **Donor:** Government of Kenya

Positions held: Team Leader/ Irrigation Engineer

Location: Kisumu County, Kenya

Main Project Features:

- Review design report and drawing for Lot 1 of the project.
- Prepare Technical designs, drawing and bid document preparation.

Activities Performed:

- Responsibility for study activities in the home office (Nairobi) and in the field,

- Co-ordinate all inputs by the key professional staff,
- Coordinate the preparation of all deliverables
- Responsible for undertaking
- Undertook review of the work: design and drawings, lead the design team in fields survey and assessment, design work, drawings and preparation of bid document for irrigation development option: 10m high concrete weir, water intake complete with the associated irrigation infrastructures (pipes, canals, canal structures and service road) for the Ahero and West Kano Irrigation Development Project, Kisumu County, Republic of Kenya.

Name of Assignment or Project: Feasibility Study, Detailed Design And Preparation Of Tender Documents For Upper Nzoia Irrigation Development Project

Year: June 2013 **To:** July 2013

Client: National Irrigation Board, Kenya

Donor: Government of Kenya

Positions held: Team Leader/ Irrigation Engineer Location: Bungoma County, Kenya Main

Project Features:

- Scoping studies for scheme evaluation.
- Field reconnaissance work –coordination.
- Technical Feasibility Study Report preparation.
- Field survey work –coordination.
- Technical designs, drawing and bid document preparation.

Activities Performed:

- Responsibility for study activities in the home office (Nairobi) and in the field,
- Co-ordinate all inputs by the key professional staff,
- Coordinate the preparation of all deliverables
- Responsible for undertaking
- Undertook scoping study, prepared scoping reports, prepared and co-ordinate the field survey work, lead the design team in design work for Taremi Dam and Nzoia Dam, drawings and preparation of bid document for irrigation development option: 35m high concrete dam, water intake complete with the associated irrigation infrastructures (pipes, canals, canal structures and service road) for the Upper Nzoia Irrigation Development Project, Bungoma County, Republic of Kenya.
- Technical designs for dams, weirs, irrigation canals and associated water control infrastructures and scheme service roads, drawing and bid document preparation.
 1. *Dams Rehabilitation and extension of the irrigable land: five (5) earth dam (15m, 12m, 10m, 8m and 6m high) complete with the associated irrigation infrastructures (canals, canal structures and service road) to irrigate 1,215ha.*
 2. *River diversion weirs Rehabilitation and extension of the irrigable land: one (1) weir complete with the associated irrigation infrastructures (canals, canal structures and service road) to irrigate 200ha*
 3. *New Dam Design and improvement of existing irrigation: eleven (11) earth dam (28m, 3 number 18m, 2 number 15m, 3 number 12m, 10m and 7m high) complete with the associated irrigation infrastructures (canals, canal structures and service road) to irrigate 3,600ha.*
 4. *New weir Design and improvement of existing irrigation: nine (9) earth dam complete with the associated irrigation infrastructures (canals, canal structures and*

service road) to irrigate 2,440ha.

5. *Overseeing and performing as needed, technical assessments for community infrastructure projects to enable separation of community input and what's requires sub-contracting due to need for specialized skills or equipment's.*
6. *Supporting community based and DASIP staff monitoring of progress and quality of work of DASIP sub-contractors with respect to contract deadlines and deliverables.*
7. *A total of 7,455ha was brought mainly under paddy irrigation*

Name of Assignment or Project: Renovation of the Hashi Fuel Depot

Year: 2008

Location: Mombasa City, Kenya Client : Hashi Energy Main Project Features:

- Technical designs work for the floating roof tanks, Tank farm assessment.
- Projects implementation- Construction and installations.

Positions held: Team Leader / Civil Structural Engineer

Activities Performed:

- Assessment of the access roads, Fuel Storage Tanks (conical fixed roof, floating roofs and walls), Tank farm bundle earth walls.
- Rehabilitating by rebuilding damaged bundle walls, vertical tanks walls, fixed conical roof floating roofs and fuel pipe works complete with all fittings installation.
- Planning and co-ordination of workshop and site works on earth bundle walls, fuel lines and vertical and horizontal tanks renovation.
- Planning and design oversight, in consultation with HASHI program counterparts, Head Office, and local staff, of small scale infrastructure projects including new construction and rehabilitation works requirement for proper project implementations.
- Provide expertise on construction sites as an Inspection Engineer. Monitor the Construction Sub Contractor's production or performance progress and notify subcontractor in writing of deficiencies observed during surveillance, and direct appropriate action to effect correction. Record and report to the Head Office and HASHI Operations Manager incidents of design errors, delays or problems.
- Monitoring quality and quantity of construction materials and equipment and verifying against contract terms, inventory reports, and accepted construction standards
- Ensuring the staff and the community works in a harmonic relationship.
- Logistic planning for the supply.
- Creating a liaison with HASHI on any security update and ensuring construction staff adhere to the strict site security rules due to nature of site: petroleum depot.

Name of Assignment or Project: Design and Construction Supervision of Independent Power Station

Year: 2008

Location: Kenya

Client: Kenya Power & Lighting Company

Main Project Features:

- Technical designs work for the Power Station.
- Water source assessment, design and construction.
- Water Storage and reticulation design and construction.

Positions held: Team Leader / Civil Structural Engineer

Activities Performed:

- Design of Independent Power Station (Water and Fuel supply reticulation line and storage tanks, sewer, buildings and access roads) for Baragoi, Mfangano Island, Laisamis, Lokichogio, Lowdar Extension Phase, Kiunga and North Horr.
- Planning and design oversight, in consultation with KPLC program counterparts, Head Office, and local staff, of small scale infrastructure projects including new construction and rehabilitation works requirement for proper project implementations.
- Overseeing and performing as needed, technical assessments for community infrastructure projects to ensure smooth inter-relation of the contractor and community component.
- Logistic planning for the supply due to controlled movement within the entire zone of operation.

Name of Assignment or Project: Design and Construction Supervision of Independent Power Station

Year: 2007

Location: Wajir County, Kenya

Client: Kenya Power & Lighting Company

Main Project Features:

- Technical designs work for the station.
- Water source assessment, design and construction.
- Projects implementation performance

Positions held: Team Leader / Civil Structural Engineer

Activities Performed:

- Design and Construction Supervision of Independent Power Station (Water {6km of 3" diameter pipes} and Fuel supply reticulation line and storage tanks {ground and elevated}, sewer, buildings and access roads) for Habaswein, Wajir South.
- Planning and design oversight, in consultation with KPLC program counterparts, Head Office, and local staff, of small scale infrastructure projects including new construction and rehabilitation works requirement for proper project implementations.
- Overseeing and performing as needed, technical assessments for community infrastructure projects to ensure smooth inter-relation of the contractor and community component.
- Provide expertise on construction sites as an Inspection Engineer. Monitor the Construction Sub Contractor's production or performance progress and notify subcontractor in writing of deficiencies observed during surveillance, and direct appropriate action to effect correction. Record and report to the Head Office and KPLC Operations Manager incidents of design errors, delays or problems.
- Monitoring quality and quantity of construction materials and equipment and verifying against contract terms, inventory reports, and accepted construction standards
- Ensuring the staff and the community works in a harmonic relationship.
- Logistic planning for the supply due to controlled movement within the entire zone of operation.
- Creating a liaison with KPLC on any security update and ensuring construction staff adhere to the strict site security rules.

Name of Assignment or Project: GOSS New Site Barracks

Year: 2006

Location: New Site, South of Sudan Client: Government of South of Sudan Main Project Features:

- Technical designs work for the Barrack., Water source assessment
- Projects implementation performance

Positions held: Team Leader / Civil Structural Engineer

Activities Performed:

- Design and Construction Supervision of Godowns, sanitation blocks, water line reticulation network, sewer line and septic tanks, access roads, Rain water harvesting, earth dam design for water storage, Generator supply and installation plus fuel storage and service bay.
- Planning and design oversight, in consultation with GOSS program counterparts, Head Office, and local staff, of small scale infrastructure projects including new construction and rehabilitation works requirement for proper project implementations.
- Overseeing and performing as needed, technical assessments for community infrastructure projects to ensure smooth inter-relation of the contractor and community component.
- Provide expertise on construction sites as an Inspection Engineer. Monitor the Construction Sub Contractor's production or performance progress and notify subcontractor in writing of deficiencies observed during surveillance, and direct appropriate action to effect correction. Record and report to the Head Office and GOSS Operations Manager incidents of design errors, delays or problems.
- Monitoring quality and quantity of construction materials and equipment and verifying against contract terms, inventory reports, and accepted construction standards
- Ensuring the staff and the community works in a harmonic relationship noting the civil wars were still on-going.
- Logistic planning for the supply due to controlled movement within the entire zone of operation noting that almost all construction materials were imported.

Name of Assignment or Project: Facility water storage and reticulation.

Year: 2001 To: 2015

Location: Kenya

Client: UNHCR /Several Institutions/ Individuals

Main Project Features:

- Technical designs.
- Water source assessment(Boreholes/ River)
- Water distribution network {pipe diameters varying in type [GRP, uPVC, GI and Epoxy painted and size- 300mm diameter to 25mm diameter}/ points (water kiosks)
- Projects implementation performance

Positions held: Civil Structural Engineer

- Activities Performed: **UNHCR / Several Institutions/ Individuals:** Design and Construction Supervision of borehole pumping units complete with associated water reticulation pipework, elevated/ and ground water storage tanks.
- Planning and design oversight, in consultation with UHNCR program counterparts, Head Office, and local staff, of small scale infrastructure projects including new construction and rehabilitation works requirement for proper project implementations.

- Overseeing and performing as needed, technical assessments for community infrastructure projects to ensure smooth inter-relation of the contractor and community component.
- Provide expertise on construction sites as an Inspection Engineer. Monitor the Construction Sub Contractor's production or performance progress and notify subcontractor in writing of deficiencies observed during surveillance, and direct appropriate action to effect correction. Record and report to the Head Office and UNHCR Operations Manager incidents of design errors, delays or problems.
- Monitoring quality and quantity of construction materials and equipment and verifying against contract terms, inventory reports, and accepted construction standards.
- Ensuring the staff and the community works in a harmonic relationship noting the refugees are from civil wars and warring communities.
- Logistic planning for the supply due to controlled movement within the route to Dadaab and Kakuma noting that almost all construction materials were from Nairobi.
- Creating a liaison with UNHCR on any security update and ensuring construction staff adhere to the strict Camp security rules

Sarova Hotels, Fair Mount Hotel, Aga Khan Hospital, GlaxoSmithKline, EPZ, Kitengela, St Mary Hospital, KTDA factories, Green park Estate-Athi River Coca-Cola Bottling (Mwanza), Kenol Kobil among others.

- Design and Construction Supervision of borehole pumping units complete with associated water reticulation pipework, elevated/ and ground water storage tanks.
- Planning and design oversight, in consultation with the consultant and the Client program counterparts, Head Office, and local staff, of small scale infrastructure projects including new construction and rehabilitation works requirement for proper project implementations.
- Overseeing and performing as needed, technical assessments for community infrastructure projects to ensure smooth inter-relation of the contractor and community component.
- Provide expertise on construction sites as an Inspection Engineer. Monitor the Construction Sub Contractor's production or performance progress and notify subcontractor in writing of deficiencies observed during surveillance, and direct appropriate action to effect correction. Record and report to the Head Office and the Consultant on the incidents of design errors, delays or problems.
- Monitoring quality and quantity of construction materials and equipment and verifying against contract terms, inventory reports, and accepted construction standards.
- Ensuring the staff and the community works in a harmonic relationship noting the refugees are from civil wars and warring communities.
- Logistic planning for the supply.

Name of Assignment or Project: Office renovation and Factory Construction.

Year: 2000

Location: Nairobi City, Kenya

Client: Plumware/ Jantech Engineering

Main Project Features:

- Technical designs.

- Working Drawings.
- Projects implementation performance

Positions held: Civil and Structural Engineer

Activities Performed:

- Design and construction supervision of Baricho Road Plumware offices renovation works.
- Design and construction supervision of Jantech Engineering go-downs Masai road off Mombasa road. Works includes 20 m wide portal frame design, overhead crane, galvanizing kiln, septic tank and masonry works.
- Planning and design oversight, in consultation with Head Office, and local staff, of small scale infrastructure projects including new construction and rehabilitation works requirement for proper project implementations.
- Provide expertise on construction sites as an Inspection Engineer. Monitor the Construction Sub Contractor's production or performance progress and notify subcontractor in writing of deficiencies observed during surveillance, and direct appropriate action to effect correction. Record and report to the Head Office incidents of design errors, delays or problems.
- Monitoring quality and quantity of construction materials and equipment and verifying against contract terms, inventory reports, and accepted construction standards.

Name of Assignment or Project: Design, Construction and Supervision

Year: 2000

Location: Kenya

Client: Meru Water Supply Donor: JICA, Government of Japan Main Project Features:

- Technical designs review.
- Working Drawings.

Positions held: Civil/ Structural Engineer

Activities Performed:

- Works entailed design of the 2 no 200 cum and 100 cum sectional steel water tanks on 9m tower and 200 mm diameter galvanized pipe-works and associated control valves to ground level.
- Planning and design oversight, in consultation with Head Office, Sumitomo2, Spencon3 and local staff, of infrastructure projects including new construction and rehabilitation works requirement for proper project implementations.
- Provide expertise on construction sites as an Inspection Engineer. Monitor the Construction Sub Contractor's production or performance progress and notify subcontractor in writing of deficiencies observed during surveillance, and direct appropriate action to effect correction. Record and report to the Head Office and Sumitomo incidents of design errors, delays or problems.
- Monitoring quality and quantity of construction materials and equipment and verifying against contract terms, inventory reports, and accepted construction standards.

10. Contact Information:

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MUIRURI PETER MAINA CV

BIODATA

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 muirurip08@gmail.com
 <https://www.linkedin.com/in/peter-muiruri-10956856/>
 Muiruri_Peter
Gender MALE | Nationality KENYAN | Birth Place MURANG'A

EDUCATION

Sept 2006-Dec 2011: BSc. in Water Engineering, Kenyatta University, Kenya.
Sept 2016-Mar 2018: MSc. Geo-information and Remote Sensing (specializing in Water Resources and Environmental Management) - ITC-University of Twente, Netherlands.

PROFESSIONAL COURSES/CERTIFICATIONS

15th -19th Dec 2014: Completion Fluids and Reservoir Drilling Fluids School, Schlumberger, France.
19th Jun-22nd Oct 2013: Basic Mud School (Drilling Fluids School) Schlumberger, Thailand
29th-31st July 2013: Quality Health Security and Environment Level 2 (QHSE-2) course, Schlumberger, Nairobi
8th-12th July 2013: Oil Field Instruments and Testing Equipment (OFITE) repair and calibration course, Schlumberger, Tanzania

SKILLS

I am skilled in Hydrology, Oilfield Drilling Fluids, WASH programming, Borehole drilling,, GIS & Remote Sensing, Hydrological Data Collection, Modelling & Analysis, Project Planning and Management, PHAST/CHAST & CLTS, and Environmental Impact Assessment.

- Computer Skills (proficient in Microsoft Office Suite, MATLAB, HEC-RAS, SWMM, AutoCAD etc.)
- Soft Skills: Team Management and Leadership, Analytical and Negotiation Skills, Public Relations and Networking

PROFESSIONAL MEMBERSHIP

- ✓ Member, Hydrological Society of Kenya (HSK, 2018), registered hydrologist.
- ✓ EIA Lead Expert -Registration number **2879**
- ✓ Graduate Engineer-Engineers Registration Board

PROFESSIONAL AND RESEARCH WORK DONE

Nov 2020: Hydrological analysis for the proposed Kyeleni Earth Dam, in Matungulu Ward, Machakos County.
Mar-Sept 2020: The site engineer for Kyelenge General Contractors Ltd during the implementation of Sereolipi and Ntabasi Boreholes Rehabilitation/improvement works in Samburu East and Rehabilitation & Desilting of Ngilai Water Pan in Samburu North (Baragoi) for ACTED Kenya: *International Tender Ref. No. T/24DUF/47F/BRH/07-02-2020/002*. Ranked the **best** of the 4 contractors engaged; for efficiency, workmanship and responsiveness.

Jan-Feb 2020:	Hydrological analysis for the proposed Miangeni Earth Dam, in Kivaa Ward, Machakos County.
July-Nov 2019:	Project Supervision during construction of the World Bank/GoK-funded Ikalaasa and Kitambaasye earth dams in Machakos County.
April 2019:	Hydrological design for the proposed Kitambaasye small earth dam in Matungulu, Machakos County and for Kilaatu water pan in Machakos County.
January 2019:	Part of the team of engineers affiliated to Agile Consulting Ltd engaged to develop a technical and financial proposal to the Ministry of Water and Sanitation for provision of CONSULTINGSERVICES FOR ATHI RIVER BASIN HYDROLOGICAL MODELLING of Thwake Dam.
May 2018-date:	Participated in developing Progressive Cavity Pump (PCP) training manual and sourced the PCP trainers for KIPYA Africa Drilling Solutions (May-July 2018). Currently involved in developing a geo-referenced national borehole repository (Borehole Map of Kenya) for WRA and KIPYA. https://boreholes.cawaters.com
2017-March 2018:	Thesis research: GIS and remote sensing application for drought monitoring and assessment (https://webapps.itc.utwente.nl/librarywww/papers_2018/msc/wrem/muiruri.pdf)
2013-2016:	Tullow Oil BV oilfield exploration and appraisal program, Schlumberger Drilling Fluids Engineer
Sept 2011- July 2012:	Implemented OXFAM WASH programs in Dadaab and Wajir.
May – August 2011:	Part of the SWAS Water Surveys team engaged to carry out geotechnical investigations for the proposed KenGen's Kenya Coal at Zowerani, Kilifi. Main duties included collection of field data and assisting the lead geologist in interpretation of findings for report compilation. Drilling supervision- Kikuyu Springs Observation Wells (WRMA) Borehole Environmental Impact Assessment and Audit for private and NGOs among them the Vice-President's Residence Borehole in Karen, Nairobi.
April-July 2011:	Borehole drilling, lined water pans, sand dams and river barrages development in Wamba, Samburu and Isiolo counties. This included design, implementation, community awareness, and the social/environmental impact assessment. EIA Reports commissioned: proposed Sabache Spring intake works for Learata Community, Samburu East, Maragua engineered landfill, Lmisysogoi, Nkototo Arus, Sai Office Supplies Ltd boreholes, among many others.
April 2011:	Unpublished undergraduate project presented to Kenyatta University: Investigating effects of fertilizers on the water quality of River Sabasaba, in Murang'a. The study sought to investigate what impacts use of nitrogenous and phosphoric fertilizers in farming had on the lifeline.
Jan – Feb 2011:	Part of the SWAS Water Surveys team engaged to carry out geotechnical studies for the Tiamin Base dam-site in Mwakitau, Ukunda and the Mwakirunge landfill in Coast Province using the Lund Imaging and coring. Roles: supervising and training semi-skilled/unskilled personnel, equipment handling, and survey set up, data collection, interpretation and reporting.
November 2010:	Developed a GoK Funded Pilot Project proposal dubbed Viability of constructing plastic lined water pans in the Machakos North County Households for Mr. Noel Bwibo- Urban and Regional Planner working with the City Council of Nairobi.

■ WORK EXPERIENCE

June 2019- Date: Technical Director, Bipem Engineering and Consultancy Ltd (Start-up)

- Providing technical leadership and direction in oil and gas, Surface & Catchment Hydrology, integrated water resources management, GIS and remote sensing, Climate Change, DDR and Environmental Impact Assessment projects.
- Developing technical design drawings, technical specifications, feasibilities, bill of quantities and EIA.
- Spearheading operations and aligning near-term and future growth prospects to the company objectives and vision
- Identifying and seeking new viable business ventures
- Ensuring company ethos are embraced for better service delivery
- Initiating and participating in CSR projects.

Jul 2018 - Mar 2019: Field Engineer, Quality Experts Ltd.

- Coordination of field teams, project management and reporting.
- Successfully managed and coordinated field experts engaged to inspect 9 tanks and 1400km gas pipeline to API 653 and API 570 NDT standards respectively for Total Energies, Eritrea Massawa Terminal.

Mar 2013 - Mar 2018: Drilling Fluids Engineer, Schlumberger

- Formulating drilling mud and optimizing performance to mitigate against hole problems: stuck pipe, lost circulation, lost time incidents (LTI), non-productive time (NPT) and HSE incidents.
- Drilled 10 wildcats and developed 5 work-over wells in Kenya, incident-free.
- Cofound and team-led the Nairobi loss prevention team (LPT) in 2013-15 and was instrumental to organizing on-location health-safety and fire-fighting trainings for the new-hires.
- Represented the project engineer in company and client meetings and was the focal person during Tullow Oil project closure in the last quarter of 2016
- Inventory management, tracking and reporting: Successfully fast-tracked the transfer of over \$10million worth drilling fluids inventory to Mozambique and Tanzania for the SASOL and STATOIL projects.
- Running ONETRAX, Virtual hydraulics (VCCP) and displacement programs

Jan 2013 - Mar 2013: Program Officer-WATER, SANITATION & HYGIENE (WASH), Feed the Children

- Facilitated timely and quality implementation of Water, Sanitation and Hygiene projects in Turkana, Samburu and Kajiado as per the project plan and coordinated and networked with the Government line ministries and other stakeholders to ensure effective implementation of WASH projects in adherence to WHO and SPHERE standards.
- Contributed in writing proposals and relief intervention plans for the Samburu and Turkana programs

Sep 2011 to Jul 2012: Public Health Engineer Assistant-WATER, SANITATION & HYGIENE (WASH), OXFAM

- Facilitated timely construction of emergency WASH infrastructure in Daadab Refugee Complex following the 2011/12 Somali refugee influx
- Supervised and coordinated contractors and implementing partners (WASDA, DPA and ALDEF) during the rehabilitation and construction of WASH facilities in 23 communities and schools in Wajir under the AusAID/DEC funded Wajir Early Recovery Program. It entailed: drilling and equipping 4 new boreholes, rehabilitation of 7 old boreholes, construction and rehab of generator houses, mini-water kiosks, water troughs, raised tank platforms, masonry tanks, ferro-cement tanks, VIP latrines and compost pits.
- Successfully organized a technical training to 23 pump operators (Rapid Response Team) to troubleshoot pumps, generators and respond to borehole breakdowns.
- In liaison with the PHP team, designed and conducted two PHAST and CHAST trainings. Behavioral change success rate of approximately 55%. Also designed and tested hygiene messages for hygiene promotion in the form of snake and ladders, talk shows, brochures, posters and banners.

Sept 2010-Sept 2011: Geologist Assistant, SWAS Water Surveys Ltd

- Conduct geotechnical and hydro-geological surveys, Aquifer mapping and borehole siting
- Supervision of borehole drilling and equipping works
- Developing technical reports, proposals and designs
- Developing work plans and project budgets.
- Environmental Impact Assessment (EIA)/Audit

May- Sept 2010: Industrial attachment, Insta-Pumps Engineering Ltd

- Field ground water exploration and borehole site investigation, interpretation and preparation of geologists and compiling subsequent hydro-geological report
- Preparation of test-pumping and borehole completion reports
- Preparation of tender documents, work plans and bill of quantities

May – Aug2007: Intern, Maragwa Water and Sewerage Company

- Provided support in implementation of water and sanitation projects (such as construction, rehabilitation and improvement of water supply and sewer facilities) to ensure efficiency in billing, water supply and record keeping.
- Oversaw the laying of 80km High Pressure Pipe (HDPE) pipeline for Karia water project

■ AWARDS/SCHOLARSHIPS

1. NUFFIC Netherlands Fellowship Program (NFP) Scholarship (2016/2018), funded the MSc. program

■ LANGUAGE SKILLS

Fluent in spoken and written

ENGLISH & SWAHILI

■ DIGITAL SKILLS

Microsoft Office: Word, Excel, Power Point, Publisher, Outlook | Computer Programming (MATLAB, STATA and R) | GIS and Remote Sensing Software (ArcGIS, Q-GIS, ILWIS, ERDAS, ENVI, ESA SNAP| Project Planning and Management (MS Project) | Team-work oriented | Organizational and planning skills | Decision-making Good listener and communicator | Written and Verbal skills | Critical thinking | Presenting | Strategic Planning Research and analytical skills.

GLADYS NDILA MUNYAO

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BIO-DATA

NATIONALITY: KENYAN

ID: NUMBER: 25495756

GENDER: FEMALE

LANGUAGES: ENGLISH AND KISWAHILI

CAREER OBJECTIVE

To fully exploit my potential and strengths by serving my employer and clients to the best of my ability, knowledge and will.

EDUCATION BACKGROUND

	NIBS COLLEGE
March 2013- June 2014:	<i>Diploma in Purchase and Supply Management</i>
	KIMS COMPUTER COLLEGE
Feb 2009-April 2009:	<i>Computer proficiency Certificate</i>
Feb 2002-Nov 2006:	TAWA MIXED SEC SCH
	<i>KCSE Certificate</i>

PROFESSIONAL SKILL SET

- Supply chain management
- Stores management and stores control
- Customer service and Negotiation
- Purchasing strategies
- Logistics and Operations chain management
- International purchasing strategies
- Microsoft office applications
- Valid driving license

SKILLS AND ABILITY

- Ability to work independently and make informed decisions
- Team player
- Procurement ethics and Laws knowledge
- E-procurement procedures knowledge
- Logistics and Operations chain management

WORK EXPERIENCE

June 2018 to Date: Procurement Officer- Bipem Engineering and Consultancy Ltd.

- Preparation of tender bids
- Process purchase requisitions / orders within purchasing
- Supplier onboarding and verification
- Transport Management Systems: Systems scoping, procurement and implementation for integrated fleet, fuel, tracking and planning/scheduling management
- Sales and marketing-negotiating with new clients and exploring new business markets for forklift services and logistics

Jan 2015 to Mar 2018: Procurement and Sourcing Officer- Kenya Toner and Ink Suppliers Ltd

Roles:

- Sourcing hi-tech printer ink cartridges and toners from approved dealers and stockists worldwide
- Warehouse management:-maintaining the inventory through stock monitoring and reconciliation
- Supplier onboarding and verification
- Preparation of tender bids and procurement
- Inventory management and procurement
- Process purchase requisitions / orders
- Assist in the development of specifications for equipment, materials and services to be purchased
- Administer contract performance including delivery, receipt, warranty, damages and insurance
- Develop and maintain constructive and cooperative working relationships with colleagues and management

July 2014 to Jan 2015: Requisition Officer- Mersons Communication Ltd.

Key responsibilities

- Warehouse management:-maintaining the inventory through stock monitoring and reconciliation
- Preparing requisite dispatch and delivery documents prior to dispatch or receiving cargo
- Logistics management
- Confirming placed orders with the departmental heads to avoid over-stocking
- Distributing circulars to the departmental heads stating policies for order placements
- Procuring the best possible price for all minor and major purchases
- Negotiating on quality and price with vendors

2009-2011: STANMATT SOKO SUPERMARKET ATTENDANT

Key responsibilities

- Conducting customer satisfaction surveys
- Forwarding any stock outs for re-ordering
- Assisting customers
- Following the made promises to meet customer needs and expectations

2007-2009: Receptionist- JACANA INTERIOR DESIGNS

Key responsibilities

- Scheduling and confirming appointments and follow-up visits
- Clerical duties like typing memos, reports, letters, correspondence and other documents as assigned
- Accepting incoming deliveries and direct them appropriately as well as assist with office mailings and sorting
- Perform responsibilities of preparing and maintaining meeting diary, electronically as well as manually
- Note written and verbal messages and ensure delivery to the intended recipient(s)
- Answer questions about the company and providing callers with address, directions, and other information

REFEREES

1. Grace Wangui Muchiri
Financial controller
Kenya Toner and Ink Suppliers
Mobile: +254 715 232 865
2. Mrs. Beth Wangui Kahari
Accountant manager
Stanmatt Soko Supermarket
Mobile: +254 722 239 203
3. Mr. Peter Muiruri
Technical Director
Bipem Engineering & Consultancy
Email: Muirurip08@gmail.com

ANNEX 2: AUDITED FINANCIAL ACCOUNTS: 2021-2018.

